R. E. Kass, L. Tierney, and J. B. Kadane. The validity of posterior expansions based on Laplace's method. In S. Geisser, J. S. Hodges, S. J. Press, and A. Zellner, editors, *Bayesian and Likelihood Methods in Statistics and Econometrics: Essays in Honor of George A. Barnard*, volume 1, pages 473–488. Elsevier, 1990.

J. K. Kruschke and T. M. Liddell. The Bayesian New Statistics: Hypothesis testing, estimation, meta–analysis, and power analysis from a Bayesian perspective. *Psychonomic Bulletin & Review*, 25:178–206, 2018.

P.-S. Laplace. Memoir on the probability of the causes of events. *Statistical Science*, 1:364–378, 1774/1986.

A. Ly, A. J. Verhagen, and E.-J. Wagenmakers. An evaluation of alternative methods for testing hypotheses, from the perspective of Harold Jeffreys. *Journal of Mathematical Psychology*, 72:43–55, 2016a.

A. Ly, A. J. Verhagen, and E.-J. Wagenmakers. Harold Jeffreys's default Bayes factor hypothesis tests: Explanation, extension, and application in psychology. *Journal of Mathematical Psychology*, 72:19–32, 2016b.

A. Ly, M. Marsman, A. J. Verhagen, R. P. P. P. Grasman, and E.-J. Wagenmakers. A tutorial on Fisher information. *Journal of Mathematical Psychology*, 80:40–55, 2017.

A. Ly, A. R. Komarlu Narendra Gupta, A. Etz, M. Marsman, Q. F. Gronau, and E.-J. Wagenmakers. Bayesian reanalyses from summary statistics and the strength of statistical evidence. *Advances in Methods and Practices in Psychological Science*, 1(3):367–374, 2018. doi: 10.1177/2515245918779348.

A. Ly, A. Stefan, J. van Doorn, F. Dablander, D. van den Bergh, A. Sarafoglou, Š. Kucharskỳ, K. Derks, Q. F. Gronau, A. R. Komarlu Narendra Gupta, U. Boehm, E.-J. van Kesteren, M. Hinne, D. Matzke, M. Marsman, and E.-J. Wagenmakers. The Bayesian methodology of Sir Harold Jeffreys as a practical alternative to the p-value hypothesis test. *Computational Brain & Behavior*, 3(2):153–161, 2020.

P. McCullagh. *Tensor methods in statistics*. Courier Dover Publications, 2018.

T. J. Mitchell and J. J. Beauchamp. Bayesian variable selection in linear regression. *Journal of the American Statistical Association*, 83:1023–1032, 1988.

R. D. Morey and J. N. Rouder. Bayes factor approaches for testing interval null hypotheses. *Psychological Methods*, 16:406–419, 2011.

R. D. Morey and J. N. Rouder. BayesFactor 0.9.12-4.2. Comprehensive R Archive Network, 2018. URL http://cran.r-project.org/web/packages/BayesFactor/index.html.

S. Pawel and L. Held. The sceptical Bayes factor for the assessment of replication success. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, in press. doi: https://doi.org/10.1111/rssb.12491. URL https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/rssb.12491.

J. N. Rouder, P. L. Speckman, D. Sun, R. D. Morey, and G. Iverson. Bayesian t tests for accepting and rejecting the null hypothesis. *Psychonomic Bulletin & Review*, 16:225–237, 2009.

J. Rousseau. Approximating interval hypothesis: p-values and Bayes factors. In J. Bernardo, M. J. Bayarri, J. O. Berger, A. P. Dawid, D. Heckerman, A. Smith, and M. West, editors, *Bayesian Statistics 8: Proceedings of the Eighth Valencia International Meeting June 2–6, 2006*, volume 8, pages 417–452. Oxford University Press, 2007.

G. Shafer and V. Vovk. *Game-Theoretic Foundations for Probability and Finance*, volume 455. John Wiley & Sons, 2019.

J. W. Tukey. The philosophy of multiple comparisons. *Statistical Science*, 6: 100–116, 1991.

J. W. Tukey. Controlling the proportion of false discoveries for multiple comparisons: Future directions. In V. S. L. Williams, L. V. Jones, and I. Olkin, editors, *Perspectives on Statistics for Educational Research: Proceedings of a Workshop*, pages 6–9, Research Triangle Park, NC, 1995. National Institute of Statistical Sciences.

A. W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.

I. Verdinelli and L. Wasserman. Computing Bayes factors using a generalization of the Savage–Dickey density ratio. *Journal of the American Statistical Association*, 90:614–618, 1995.

D. Wrinch and H. Jeffreys. On certain fundamental principles of scientific inquiry. *Philosophical Magazine*, 42:369–390, 1921.

A Laplace Approximation

The Laplace approximation uses a (multivariate) Taylor expansion for which we introduce notation. Let $h : \Theta \subset \mathbb{R}^p \to \mathbb{R}$, i.e., $h(\theta) = -\frac{1}{n}\sum_{i=1}^n \log f(y_i \mid \theta)$, and we write $\hat{\theta}$ for the point in its domain where h takes its global minimum. Furthermore, we use subscripts to denote partial derivatives, whereas superscripts refer to components of a vector, or more generally an array. For instance, $\pi_a = \frac{\partial}{\partial \theta_a}\pi(\hat{\theta})$ refers to the a-th component of the vector of partial derivatives $[D^1\pi(\hat{\theta})]$ of the prior π evaluated at the MLE. Similarly, we write $h_{abc} = \frac{\partial^3}{\partial\theta_a\partial\theta_b\partial\theta_c}h(\hat{\theta})$ for the abc-th component of the three-dimensional array $[D^3h(\hat{\theta})]$. Hence, the number of indices in the subscript corresponds to the number of derivatives of h and the indices, each in $1, 2, \ldots, p$, provide the location of the component.

We use superscripts to refer to the component of a vector. For instance, $\tilde{q}^a = (\theta^a - \hat{\theta}^a)$ represents the a-th component of the difference vector $\tilde{q} = \theta - \hat{\theta}$, thus, equivalently $\tilde{q}^a := e_a^T\tilde{q}$, where e_a is the unit (column) vector with entry 1 at index a and zero elsewhere. Similarly, ς^{abcd} the $abcd$-th component of a four dimensional array.

Moreover, we employ Einstein's summation convention and suppress the sum whenever an index occurs in both the sub and superscript. For instance,

$$h_a\tilde{q}^a := \sum_{a=1}^p h_a\tilde{q}^a, \tag{24}$$

$$h_{abc}\tilde{q}^a\tilde{q}^b\tilde{q}^c := \sum_{a=1}^p\sum_{b=1}^p\sum_{c=1}^p h_{abc}\tilde{q}^a\tilde{q}^b\tilde{q}^c. \tag{25}$$

The former defines an inner product between the gradient of h and deviations $\tilde{q}$, whereas the $h_{abc} = [D^3h]_{abc}$ refers to the a-th row, b-th column, and c-th depth of the three-dimensional array consisting of partial derivatives of h of order three. Lastly, we use the shorthand notation

$$h_ah_b\tilde{q}^a\tilde{q}^b := \sum_a\sum_b h_ah_b\tilde{q}^a\tilde{q}^b, \tag{26}$$

to denote the nested sum which is needed for Cauchy products $(h_a\tilde{q}^a)(h_b\tilde{q}^b)$. For instance, with $d = 2$

$$(h_1\tilde{q}^1 + h_2\tilde{q}^2)(h_1\tilde{q}^1 + h_2\tilde{q}^2) = h_1\tilde{q}^1h_1\tilde{q}^1 + 2h_1\tilde{q}^1h_2\tilde{q}^2 + h_2\tilde{q}^2h_2\tilde{q}^2, \tag{27}$$

which is equivalent to

$$h_1h_1\tilde{q}^1\tilde{q}^1 + h_1h_2\tilde{q}^1\tilde{q}^2 + h_2h_1\tilde{q}^2\tilde{q}^1 + h_2h_2\tilde{q}^2\tilde{q}^2. \tag{28}$$

With these notational conventions a multivariate Taylor approximation is denoted as

$$h(\theta) = h(\hat{\theta}) + h_a\tilde{q}^a + \tfrac{h_{ab}}{2!}\tilde{q}^au^b + \tfrac{h_{abc}}{3!}\tilde{q}^a\tilde{q}^b\tilde{q}^c + \tfrac{h_{abcd}}{4!}\tilde{q}^a\tilde{q}^b\tilde{q}^c\tilde{q}^d + \mathcal{O}(|u|^5). \tag{29}$$

and note the similarity to its one-dimensional counterpart.

Theorem 3 (Laplace expansion with error term) *Let $\mathcal{P}_\Theta$ be a collection of density functions that are six times continuously differentiable in $\theta \in \Theta \subset \mathbb{R}^p$, and $\pi(\theta)$ a prior density that is four times continuously differentiable. Let $Y \overset{\text{iid}}{\sim} f(y \mid \theta)$ for certain θ, then with $\hat{\theta}$ the MLE*

$$p(y^n) = \int_\Theta f(y^n \mid \theta)\pi(\theta)\mathrm{d}\theta \tag{30}$$

$$= (\tfrac{2\pi}{n})^{\frac{p}{2}} f(y^n \mid \hat{\theta})\pi(\hat{\theta})|\hat{I}(\hat{\theta})|^{-1/2}\Big[1 + \tfrac{C^{(1)}(\hat{\theta})}{n} + \tfrac{C^{(2)}(\hat{\theta})}{n^2} + \mathcal{O}(n^{-3})\Big], \tag{31}$$